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(54) **STAGGERED THREE-DIMENSIONAL CHIP STACKING**

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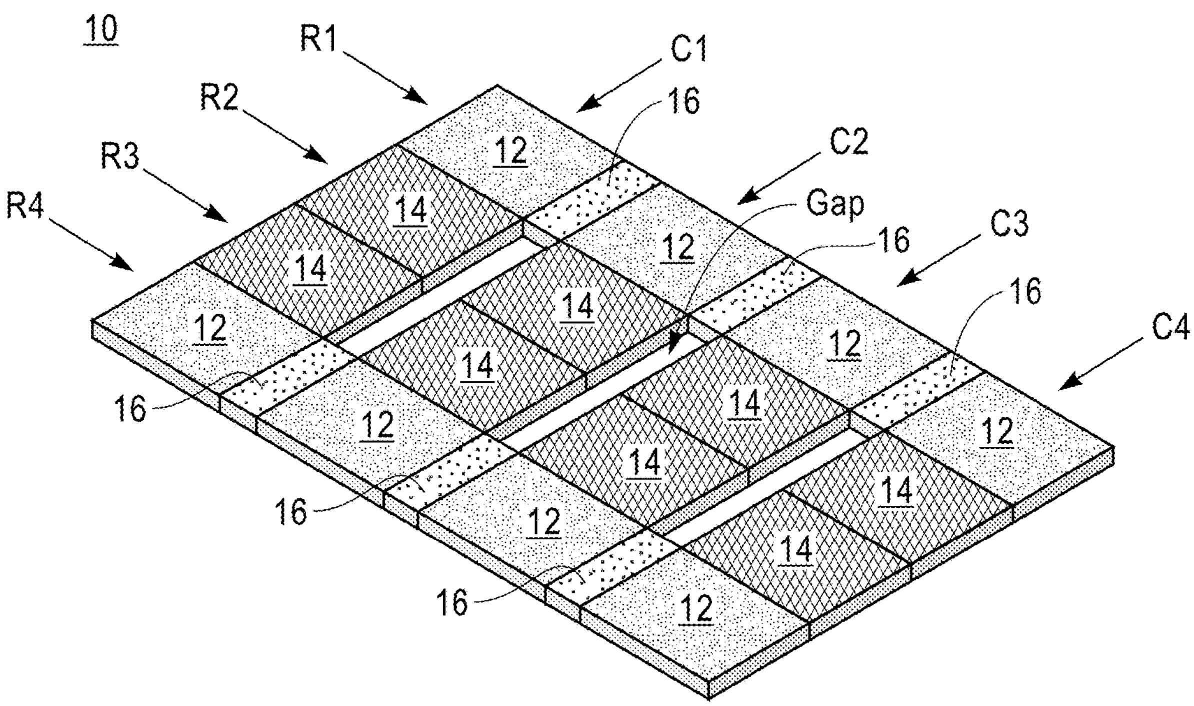
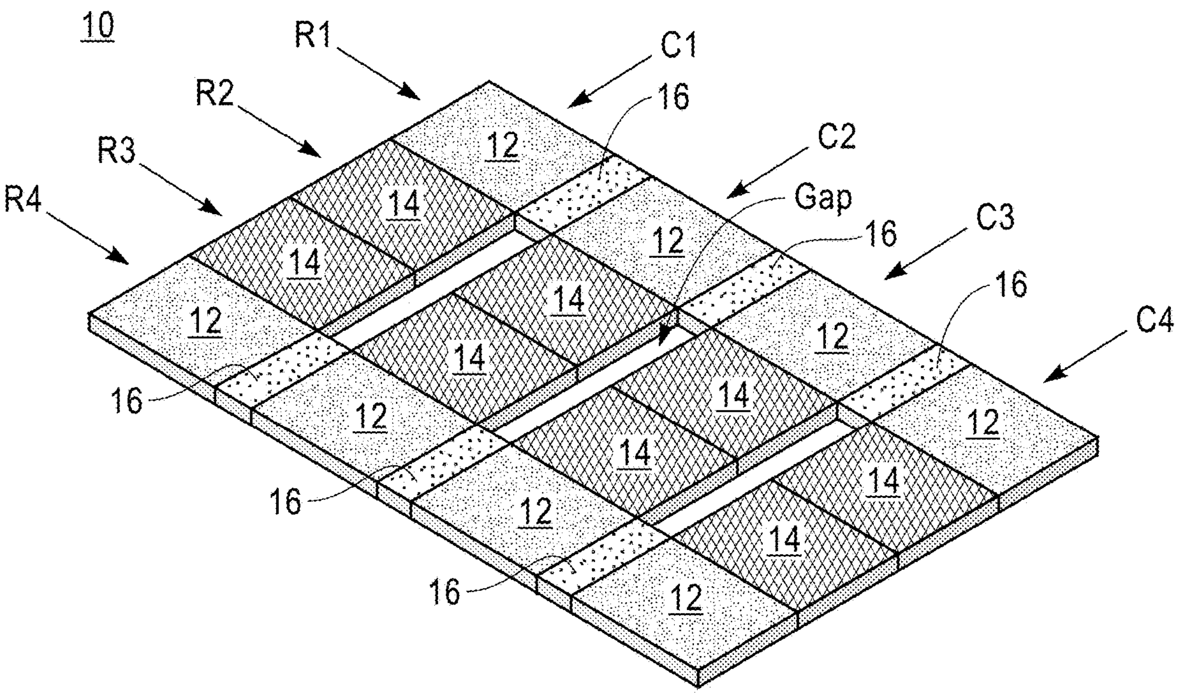
(57) **ABSTRACT**

A 3D chip stack is provided including a first chip (including core units, cache units and power elements) and a second chip (including core units, cache units and heat spreader elements) that are stacked one on top of the other in a staggered manner and a power via is present that connects one of the power elements of the first chip to one of the core units of the second chip, and a thermal via is present that connects one of the core units of the first chip to one of the heat spreader elements of the second chip.

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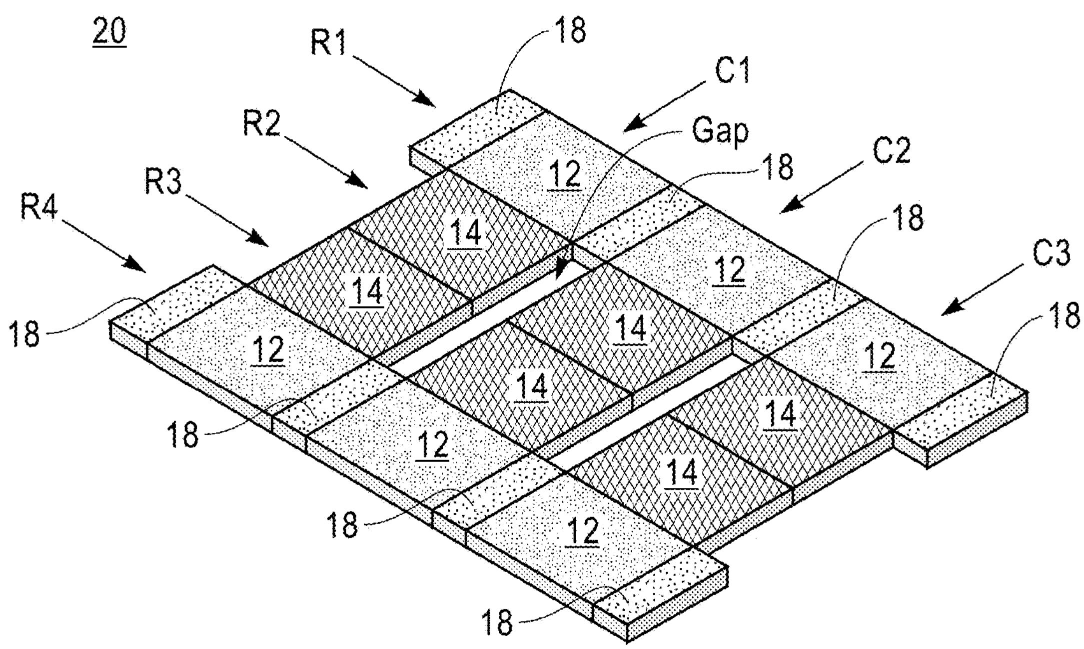
R1
R2
R3
R4
C1
C2
C3
C4
Gap
12
14
16

FIG. 1


20
R1
R2
R3
R4
C1
C2
C3
Gap
12
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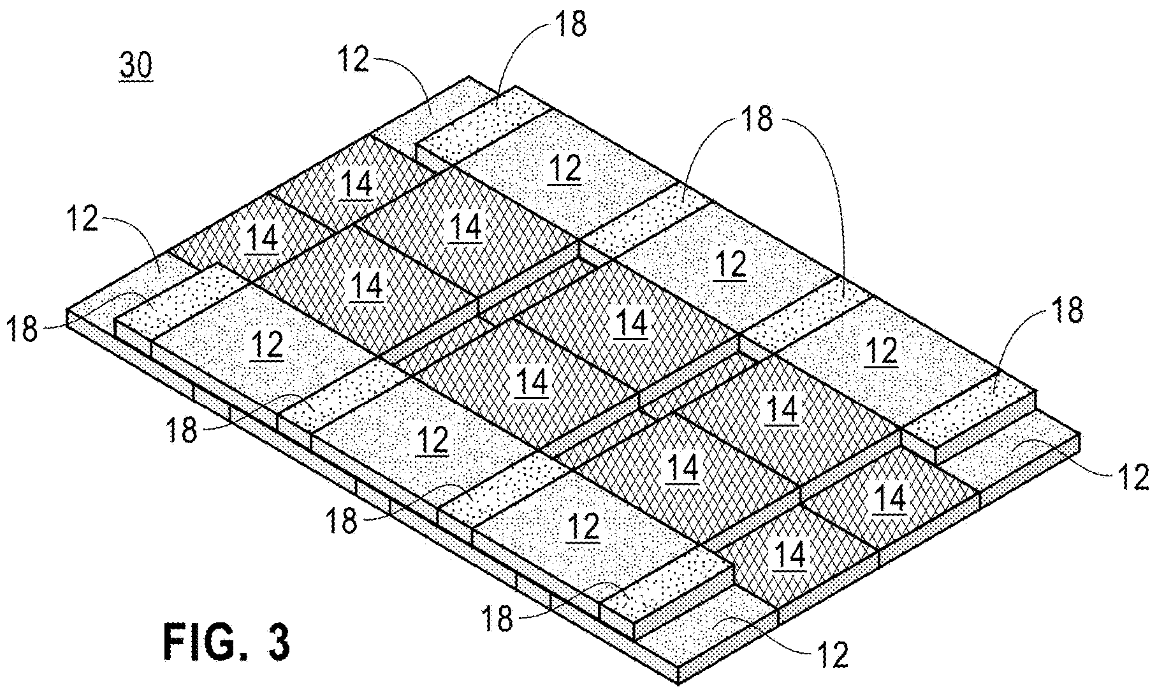
18

FIG. 2


30
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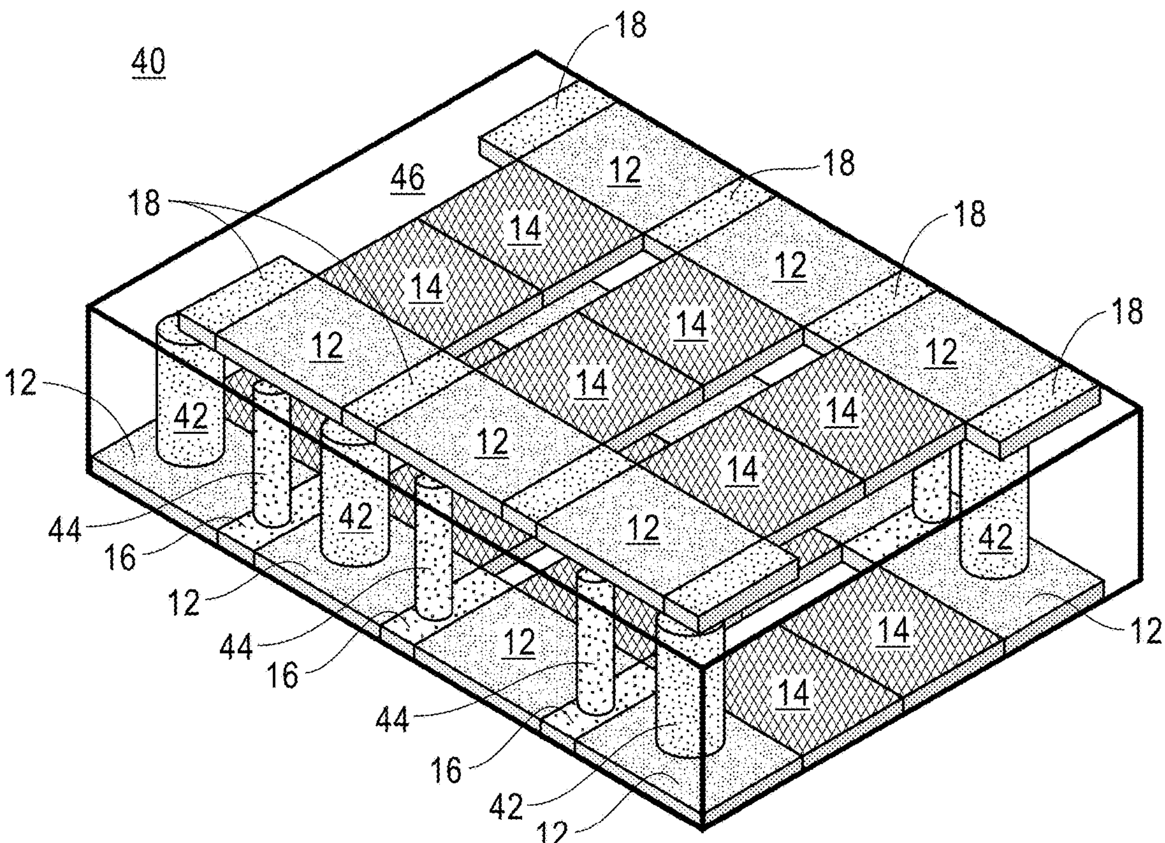
18
12

FIG. 3


40
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12
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46
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42
12

FIG. 4

STAGGERED THREE-DIMENSIONAL CHIP STACKING

BACKGROUND

[0001] The present application relates to semiconductor technology, and more particularly to a three-dimensional (3D) chip stack structure including a first chip and a second chip that are stacked in a staggered manner.

[0002] 3D chip stacking is a technique used to manufacture MOS (metal-oxide semiconductor) integrated circuits (ICs) by stacking a plurality of ICs and interconnecting them vertically using, for instance, through-silicon vias (TSVs) or Cu—Cu connections. This technique allows the ICs to behave as a single device, which can lead to performance improvements at reduced power and a smaller footprint than conventional two-dimensional processes.

SUMMARY

[0003] A 3D chip stack is provided including a first chip (including core units, cache units and power elements) and a second chip (including core units, cache units and heat spreader elements) that are stacked one on top of the other in a staggered manner and a power via is present that connects one of the power elements of the first chip to one of the core units of the second chip, and a thermal via is present that connects one of the core units of the first chip to one of the heat spreader elements of the second chip.

[0004] In one embodiment of the present application, the 3D chip stack includes a first chip including a first number of core units and a first number of cache units, and having power elements located between each neighboring core unit of the first chip. The 3D chip stack also includes a second chip including a second number of core units and a second number of cache units, and having a heat spreader element located on at least one of the core units of the second chip. In the present application, the first chip and the second chip are stacked one on top of the other in a staggered manner. The 3D chip stack of the present application further includes a power via connecting one of the power elements of the first chip to one of the core units of the second chip, and a thermal via connecting one of the core units of the first chip to one of the heat spreader elements of the second chip. In the 3D chip stack, the heat spreader element is placed with reference to thermal spots on the first chip and connected to a bottom surface where the thermal spot are prominent by the thermal vias.

[0005] In another embodiment of the present application, the 3D chip stack includes a first chip including a first number of core units and a first number of cache units, and having power elements located between each neighboring core unit of the first chip. The 3D chip stack also includes a second chip including a second number of core units and a second number of cache units, and having a heat spreader element located on at least one of the core units of the second chip. In the present application, the first chip and the second chip are stacked one on top of the other in a staggered manner, and the core units of the first chip are located on the periphery of the first chip, and the core units of the second chip are located on the periphery of the second chip. The 3D chip stack of this embodiment of the present application further includes a power via connecting one of the power elements of the first chip to one of the core units of the second chip, and a thermal via connecting one of the core units of the first chip to one of the heat spreader elements of the second chip. In the 3D chip stack, the heat spreader element is placed with reference to thermal spots on the first chip and connected to a bottom surface where the thermal spot are prominent by the thermal vias.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is an illustration of a first chip that can be employed in accordance with an embodiment of the present application.

[0007] FIG. **2** is an illustration of a second chip that can be employed in accordance with an embodiment of the present application.

[0008] FIG. **3** is an illustration showing a 3D chip stack structure in which the second chip of FIG. **2** is stacked in a staggered manner over the first chip of FIG. **1**.

[0009] FIG. **4** is an illustration showing a 3D chip stack structure in which the second chip of FIG. **2** is stacked in a staggered manner over the first chip of FIG. **1** and including power vias and thermal vias.

DETAILED DESCRIPTION

[0010] The present application will now be described in greater detail by referring to the following discussion and drawings that accompany the present application. It is noted that the drawings of the present application are provided for illustrative purposes only and, as such, the drawings are not drawn to scale. It is also noted that like and corresponding elements are referred to by like reference numerals.

[0011] In the following description, numerous specific details are set forth, such as particular structures, components, materials, dimensions, processing steps and techniques, in order to provide an understanding of the various embodiments of the present application. However, it will be appreciated by one of ordinary skill in the art that the various embodiments of the present application may be practiced without these specific details. In other instances, well-known structures or processing steps have not been described in detail in order to avoid obscuring the present application.

[0012] It will be understood that when an element as a layer, region or substrate is referred to as being “on” or “over” another element, it can be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or “directly over” another element, there are no intervening elements present. It will also be understood that when an element is referred to as being “beneath” or “under” another element, it can be directly beneath or under the other element, or intervening elements may be present. In contrast, when an element is referred to as being “directly beneath” or “directly under” another element, there are no intervening elements present.

[0013] The terms substantially, substantially similar, about, or any other term denoting functionally equivalent similarities refer to instances in which the difference in length, height, or orientation conveys no practical difference between the definite recitation (e.g., the phrase sans the substantially similar term), and the substantially similar variations. In one embodiment, substantial (and its derivatives) denote a difference by a generally accepted engineering or manufacturing tolerance for similar devices, up to, for example, 10% deviation in value or 10° deviation in angle.

[0014] As mentioned above, 3D chip stacking is used in the semiconductor industry to manufacture MOS ICs by stacking a plurality of ICs one atop the other and thereafter interconnecting them vertically. Existing 3D chip stacking techniques focus on consistent monolithic unit over unit solutions. For example, in conventional 3D chip stacking cache units are stacked over cache units, and logic units are stacked over logic units. Also, and in existing 3D chip stacking techniques, 3D thermal solution focus on restricting units on an all-or-nothing basis. That is, upper hot units are not stacked over lower hot units. Further, and in existing 3D chip stacking techniques TSV solutions focus on an all or nothing solution. An integrated approach to optimize these elements across 3D layers is lacking in the semiconductor industry.

[0015] A 3D chip stack (see, for example, FIG. **4**) is provided including a first chip (including core units, cache units and power elements) and a second chip (including core units, cache units and heat spreader elements) that are stacked one on top of the other in a staggered manner and a power via is present that connects one of the power elements of the first chip to one of the core units of the second chip, and a thermal via is present that connects one of the core units of the first chip to one of the heat spreader elements of the second chip. The staggered orientation of the first chip and the second chip reduces and confines power/thermal management points.

[0016] Throughout the present application, the term "core unit" denotes a microprocessor that carriers out programs and multiple other actions on a computer. The core unit can contain a plurality of transistors including logic transistors, as well as device circuitry including I/O circuitry.

[0017] Throughout the present application, the term "cache unit" denotes a smaller, faster memory unit, which is located closer to the core unit than a computer's main memory. The cache can store copies of the data from frequently used main memory locations.

[0018] Throughout the present application, the term "power element" denotes a power rail that delivers power to adjacent core units of the first chip and one of the core units of the second chip. In some embodiments, the power element is composed of an electrically conductive power rail material (refractory or high conductivity) including, but not limited to, tungsten (W), cobalt (Co), ruthenium (Ru), copper (Cu), platinum (Pt), rhodium (Rh), or palladium (Pd).

[0019] Throughout the present application, the term "heat spreader element" denotes a component that increases the heat flow away from a hot device/area such as, for example, one of the core units of the second chip. The heat spreader element is composed of a thermally conductive material such as, for example, silicon (Si), Cu or diamond-like carbon (DLC). In the present application, the heat spreader element is positioned on each end of the core units of the second chip.

[0020] Throughout the present application, the term "power via" denotes a via that is used in the present application to deliver power from the power element present in the first chip to one of the core units that is present in the second chip. In some embodiments, the power via can be composed of an electrically conductive metal or an electrically conductive metal alloy. Examples of electrically conductive metals include, but are not limited to, W, Cu, Al, Co, Ru, Ru, molybdenum (Mo), osmium (Os), or iridium (Ir). Examples of electrically conductive metal alloys include at least one of the electrically conductive metals mentioned above. An example of an electrically conductive metal alloy that can be employed in the present application includes, but is not limited to, Cu—Al. In some embodiments of the present application, the power element and the power via can be composed of a compositionally same electrically conductive material. In other embodiments of the present application, the power element and the power via can be composed of compositionally different electrically conductive materials.

[0021] Throughout the present application, the term "thermal via" denotes a via structure that transfers heat from a core unit of the first chip to a heat spreader element that is located in the second chip. The via structure that provides the thermal via of the present application is filled with one of the thermally conductive materials mentioned above for the heat spreader element. In some embodiments of the present application, the heat spreader element and the thermal via can be composed of a compositionally same thermal conductive material. In other embodiments of the present application, the heat spreader element and the thermal via can be composed of compositionally different thermal conductive materials.

[0022] Throughout the present application, the term "staggered manner" denotes one of the first chip or the second chip is positioned over the other of the first chip or the second chip in such a manner that each core unit of the second chip is located either under or over two adjacent core units of the first chip. That is, the staggering offsets the core units of the second chip from the core units of the first chip. Embodiments include when in which the second chip is located over the first chip, or when the second chip is located under the first chip.

[0023] Referring now to FIG. **1**, there is shown a first chip **10** (or first microprocessor) that can be employed in accordance with an embodiment of the present application. The first chip **10** includes a number of core units **12** and cache units **14** as shown in FIG. **1**. In the present application, power from the first chip **10** can be delivered to the second chip **20** by means of using power vias **44** (as shown in FIG. **4**); this aspect of the present application is described in more detail herein below. The first chip **10** is an independent unit with respect to the second chip **20**. The first chip **10** can be independently tested. The first chip **10** can have size that is less than or greater than the second chip **20**.

[0024] In the illustrated embodiment, the first chip **10** includes four rows, namely R**1**, R**2**, R**3** and R**4**, and four columns, namely C**1**, C**2**, C**3** and C**4**. Notably, the illustrated first chip **10** includes a first row R**1** and a fourth row R**4** of core units **12** that are spaced apart by a power element **16**. First chip **10** also includes a second row R**2** and a third row R**3** of cache units **14**. Each cache unit **14** within the second row R**2** and the third row R**3** of the first chip **10** is spaced apart by a gap. In the illustrated first chip **10**, the cache units **14** of the first chip **10** in each column are in direct physical contact with each other. Although FIG. **1** illustrates a first chip **10** that is a 4×4 design (i.e., row×column design), the present application is not limited to a 4×4 design for the first chip **10**. Instead, the present application contemplates using a 2N×2N design for the first chip **10** in which N is an integer starting from 1, and including 2, 3, 4, 5, etc. The first chip **10** can be formed utilizing semiconductor processing techniques that are well known to those skilled in the art.

[0025] Referring now to FIG. 2, there is shown a second chip **20** (or second microprocessor) that can be employed in accordance with an embodiment of the present application. In the illustrated embodiment of the present application, the second chip **20** is designed to have less core units **12** and less cache units **14** than the first chip **10**. In other embodiments, the second chip **20** is designed to have greater core units **12** and greater cache units **14** than the first chip **10**. In the present application, power from the first chip **10** can be delivered to the second chip **20** by means of using power vias. In the illustrated embodiment of the present application, the second chip **20** includes heat spreader elements **18** located at opposing edges of each of the core units **12** which can be used to cool the second chip **20** as well as the first chip **10**; cooling of the first chip **10** is achieved by means of thermal vias **42** (as shown in FIG. **4**) which transfer the heat generated by the core units **12** of the first chip **10** into the thermal spreader elements **18** of the second chip **20**. It is noted that the present application is not limited to include heat spreader elements **18** located on opposing edges (ends) of each of the core units **12** of the second chip **20**. Instead, the present application works when heat spreader elements **18** are located on a specific core unit **12** of the second chip **20**. The second chip **20** is an independent unit with respect to the first chip **20**. The second chip **10** can be independently tested.

[0026] In the illustrated embodiment, the second chip **20** includes four rows, namely R**1**, R**2**, R**3** and R**4**, and three columns, namely C**1**, C**2** and C**3**. Notably, the illustrated second chip **20** includes a first row R**1** and a fourth row R**4** of core units **12** that are spaced apart by a heat spreader element **18**. Second chip **20** also includes a second row R**2** and a third row R**3** of cache units **14**. Each cache unit **14** within the second row R**2** and the third row R**3** of the second chip **20** is spaced apart by a gap. In the illustrated second chip **20**, the cache units **14** of the second chip **20** in each column are in direct physical contact with each other. Although FIG. **2** illustrates a second chip **20** that is a 4×3 design, the present application is not limited to a 4×3design for the second chip **20**. Instead, the present application contemplates using a 2N×2N-1 design in which Nis as defined above; note that the second chip includes one less column than the first chip **10**. In some embodiments, the second chip **20** can have a 2N×2N design and the first chip **10** can have a 2N×2N-1 design. The second chip **20** can be formed utilizing semiconductor processing techniques that are well known to those skilled in the art.

[0027] Referring now to FIG. **3**, there is shown a 3D chip stack structure **30** in which the second chip **20** of FIG. **2** is stacked in a staggered manner over the first chip **10** of FIG. **1**. FIG. **3** represents a simplistic view of a 3D chip stack structure (bonding or thermal and power vias not shown) in accordance with the present application. This simplistic view does not include the power vias and thermal vias mentioned above. FIG. **4** shows a 3D chip stack structure **40** in which the second chip **20** of FIG. **2** is stacked in a staggered manner over the first chip **10** of FIG. **1** and including power vias **44** and thermal vias **42**. FIGS. **3** and **4** represent embodiments illustrating front-to-back-stacking. In the front-to-back-stacking embodiment illustrated in FIGS. **3** and **4**, the frontside of the first chip **10** faces the backside of the second chip **20** and the power vias **44** and the thermal vias **42** provide a connection from the frontside of the first chip **10** to the backside of the second chip **20**.

[0028] In some embodiments (not shown but readily discernable from FIG. **4**), the frontside of the second chip **20** can face the backside of the first chip **10** and the power vias **44** and the thermal vias **42** provide connection of the frontside of the second chip **20** to the backside of the first chip **10**.

[0029] In still other embodiments (not shown but readily discernable from FIG. **4**), front-to-front stacking or back-to-back stacking can be employed. In the front-to-front stacking embodiment, the frontside of the first chip **10** faces the frontside of the second chip **20** and the power vias **44** and the thermal vias **42** provide connection of the frontside of the first chip **10** to the frontside of the second chip **20**. In the back-to-back stacking embodiment, the backside of the first chip **10** faces the backside of the second chip **20** and the power vias **44** and the thermal vias **42** provide connection of the backside of the first chip **10** to the backside of the second chip **20**.

[0030] As is illustrated in FIG. **4**, the 3D chip structure of the present application can be encased in dielectric material structure **46**. The dielectric material structure **46** provides electrically illustration to the 3D chip stack structure. Dielectric material structure **46** can include one or more dielectric layers that are composed of a material such as, for example, silicon oxide, silicon nitride, undoped silicate glass (USG), fluorosilicate glass (FSG), borophosphosilicate glass (BPSG), a spin-on low-k dielectric layer, a chemical vapor deposition (CVD) low-k dielectric layer, or any combination thereof. The term "low-k" as used throughout the present application denotes a dielectric material that has a dielectric constant of less than 4.0 (all dielectric constants mentioned herein are relative to a vacuum unless otherwise noted. In some embodiments, the dielectric material structure **46** can include one or more air gaps.

[0031] In the illustrated embodiment, the stacking includes positioning the second chip **20** over the first chip **10** in a staggered manner as defined above. Prior to this positioning, the power vias **44** and the thermal vias **42** are formed within one of the dielectric layers of the dielectric material structure **46**. The power vias **44** and the thermal vias **42** can be formed utilizing a damascene process or subtractive patterning process, both of which are well known to those skilled in the art. Bond pads (not shown) can be formed on top of the power vias **44** and thermal vias **42**. As mentioned above, the thermal vias **42** is formed on a core unit **12** of the first chip **10**. The number of thermal vias **42** that are formed can vary. As mentioned above, the power vias **44** is formed on a power element **16** that is located between adjacent core units **12** of the first chip **10**. After the positioning step, the second chip **20** is bonded to the dielectric layer including the power vias **44** and the thermal vias **42**. Bonding can include a thermal bonding process or any other type of bonding process. Bond pads (not shown) can be positioned on the second chip **20** and aligned with the bond pads present on the power vias **42** and thermal vias **44**.

[0032] Referring back to FIG. **4**, there is illustrated a 3D chip stack in accordance with the present application. Notably, the 3D chip stack illustrated in FIG. **4** includes first chip **10** including a first number of core units **12** and a first number of cache units **14**, and having power elements **16** located between each neighboring core unit **12** of the first chip **10**. The 3D chip stack also includes second chip **20** including a second number of core units **12** and a second number of cache units **14**, and having a heat spreader element **18** located on at least one of the core units **12** of the second chip **20**. In the illustrated embodiment, the heat spreader element **18** is located on opposing ends of each core unit **12** of the second chip **20**. In the present application, the heat spreader element **18** is placed with reference to thermal spots on the first chip **10** and connected to a bottom surface where the thermal spot are prominent by the thermal vias **42**.

[0033] In the present application, the first chip **10** and the second chip **20** are stacked one on top the other in a staggered manner. In the illustrated embodiment, the second number is less than the first number. In other embodiments, the second number is greater than the first number. The staggered orientation of the first chip **10** and the second chip **20** reduces and confines power/thermal management points. The 3D chip stack of the present application further includes power via **44** connecting one of the power elements **18** of the first chip **10** to one of the core units **12** of the second chip **20**, and thermal via **42** connecting one of the core units **12** of the first chip **10** to one of the heat spreader elements **18** of the second chip **20**.

[0034] The presence of the power via **44** allows for power delivery from the first chip **10** to the second chip **20**. Since the power via **44** is connected to one of the power elements **18** of the first chip **10**, the impact of the power via **44** on the core units **12** and cache units **14** of the first chip **10** are reduced and sometimes even eliminated. This enables a more efficient design layout for performance-critical logic. The presence of the thermal vias **42** allows for heat dissipation from the core units **12** of the first chip **10** to the heat spreader elements **18** of the second chip **20**. The thermal via **42** and thermal spreader element **18** eliminate hotspot stacking the inhibits traditional 3D chip stacking. Also, the staggered 3D chip stack illustrated in FIG. **4** provides performance and density improvements compared to conventional 3D chip stacking.

[0035] In some embodiments of the present application, the first chip **10** is located beneath the second chip **20**. In such a stacked configuration it is possible to have (a) a frontside of the first chip **10** facing a frontside of the second chip **20**, (b) a backside of the first chip **10** facing a backside of the second chip **20**, (c) a frontside of the first chip **10** facing a backside of the second chip **20**, and (d).a backside of the first chip **10** facing a frontside of the second chip **20**.

[0036] In some embodiments of the present application, the first chip **10** is located above the second chip **20**. In such a stacked configuration it is possible to have (a) a frontside of the first chip **10** facing a frontside of the second chip **20**, (b) a backside of the first chip **10** facing a backside of the second chip **20**, (c) a frontside of the first chip **10** facing a backside of the second chip **20**, and (d).a backside of the first chip **10** facing a frontside of the second chip **20**.

[0037] In some embodiments of the present application, dielectric material structure **46** is present that encases the first chip **10**, the second chip **20**, the power via **44** and the thermal via **42**. The dielectric material structure **46** provides electrically isolation within the 3D chip stacked structure.

[0038] In embodiments of the present application, the first number is different from (i.e., greater than or less than) the second number.

[0039] In embodiments of the present application, the first chip has a size that differs from (i.e., greater than or less than) a size of the second chip.

[0040] In embodiments of the present application, a gap is located between each cache unit **14** within a row of caches units of both the first chip **10** and the second chip **20**.

[0041] In embodiments of the present application, each core unit **12** of the second chip **20** lies between a pair of neighboring cache units **14** of the first chip **10**.

[0042] Again referring back to FIG. **4**, the illustrated 3D chip stack includes first chip **10** including a first number of core units **12** and a first number of cache units **14**, and having power elements **16** located between each neighboring core unit **12** of the first chip **10**. The 3D chip stack also includes second chip **20** including a second number of core units **12** and a second number of cache units **14**, and having a heat spreader element **18** located on at least one of the core units **12** of the second chip **20**; in the illustrated embodiment the heat spreader element **18** is located on opposing ends of each of the core units **12** of the second chip **20**. In the present application, the first chip **10** and the second chip **20** are stacked one on top the other in a staggered manner. The staggered orientation of the first chip **10** and the second chip **20** reduces and confines power/thermal management points. As is shown, the core units **12** of the first chip **10** are located on the periphery of the first chip **10**, and the core units **12** of the second chip **20** are located on the periphery of the second chip **20**. This aspect of the present application limits the need for forming the power vias **44** and the thermal vias **42** with functional areas of the first and second chips. The 3D chip stack further includes power via **44** connecting one of the power elements **18** of the first chip **10** to one of the core units **12** of the second chip **20**, and thermal via **42** connecting one of the core units **12** of the first chip **10** to one of the heat spreader elements **18** of the second chip **20**.

[0043] The presence of the power via **44** allows for power delivery from the first chip **10** to the second chip **20**. Since the power via **44** is connected to one of the power elements **18** of the first chip **10**, the impact of the power via **44** on the core units **12** and cache units **14** of the first chip **10** are reduced and sometimes even eliminated. This enables a more efficient design layout for performance-critical logic. The presence of the thermal vias allows for heat dissipation from the core units **12** of the first chip **10** to the heat spreader elements **18** of the second chip **20**. The thermal via **42** and thermal spreader element **18** eliminate hotspot stacking the inhibits traditional 3D chip stacking. Also, the staggered 3D chip stack illustrated in FIG. **4** provides performance and density improvements compared to conventional 3D chip stacking.

[0044] In some embodiments, in which the core units **12** of the first chip **10** are located on the periphery of the first chip **10**, and the core units **12** of the second chip **20** are located on the periphery of the second chip **20**, the first chip **10** is located beneath the second chip **20**.

[0045] In some embodiments, in which the core units **12** of the first chip **10** are located on the periphery of the first chip **10**, and the core units **12** of the second chip **20** are located on the periphery of the second chip **20**, the first chip **10** is located above the second chip **20**.

[0046] In some embodiments, in which the core units **12** of the first chip **10** are located on the periphery of the first chip **10**, and the core units **12** of the second chip **20** are located on the periphery of the second chip **20**, the structure can further include dielectric material structure **46** encasing the first chip **10**, the second chip **20**, the power via **44** and the thermal via **42**. The dielectric material structure **46** provides electrical isolation within the structure.

[0047] While the present application has been particularly shown and described with respect to preferred embodiments thereof, it will be understood by those skilled in the art that the foregoing and other changes in forms and details may be made without departing from the spirit and scope of the present application. It is therefore intended that the present application not be limited to the exact forms and details described and illustrated, but fall within the scope of the appended claims.

What is claimed is:

1. A 3D chip stack structure comprising:

a first chip comprising a first number of core units and a first number of cache units, and having power elements located between each neighboring core unit of the first chip;

a second chip comprising a second number of core unit and a second number of cache units, and having a heat spreader element located on at least one of the core units of the second chip, wherein the first chip and the second chip are stacked one on top the other in a staggered manner;

a power via connecting one of the power elements of the first chip to one of the core units of the second chip; and

a thermal via connecting one of the core units of the first chip to one of the heat spreader elements of the second chip.

2. The 3D chip stack structure of claim **1**, wherein the first chip is located beneath the second chip.

3. The 3D chip stack structure of claim **2**, wherein a frontside of the first chip faces a frontside of the second chip.

4. The 3D chip stack structure of claim **2**, wherein a backside of the first chip faces a backside of the second chip.

5. The 3D chip stack structure of claim **2**, wherein a frontside of the first chip faces a backside of the second chip.

6. The 3D chip stack structure of claim **2**, wherein a backside of the first chip faces a frontside of the second chip.

7. The 3D chip stack structure of claim **1**, wherein the first chip is located above the second chip.

8. The 3D chip stack structure of claim **7**, wherein a frontside of the first chip faces a frontside of the second chip.

9. The 3D chip stack structure of claim **7**, wherein a backside of the first chip faces a backside of the second chip.

10. The 3D chip stack structure of claim **7**, wherein a frontside of the first chip faces a backside of the second chip.

11. The 3D chip stack structure of claim **7**, wherein a backside of the first chip faces a frontside of the second chip.

12. The 3D chip stack structure of claim **1**, further comprising a dielectric material structure encasing the first chip, the second chip, the power via and the thermal via.

13. The 3D chip stack structure of claim **1**, wherein the first number is different from the second number.

14. The 3D chip stack structure of claim **1**, wherein the first chip has a size that is different from a size of the second chip.

15. The 3D chip stack structure of claim **1**, further comprising a gap between each cache unit within a row of caches units of both the first chip and the second chip.

16. The 3D chip stack structure of claim **1**, wherein each core unit of the second chip lies between a pair of neighboring cache units of the first chip.

17. A 3D chip stack structure comprising:

a first chip comprising a first number of core units and a first number of cache units, and having power elements located between each neighboring core unit of the first chip;

a second chip comprising a second number of core unit and a second number of cache units, and having a heat spreader element located on at least one of the core units of the second chip, wherein the first chip and the second chip are stacked one on top the other in a staggered manner, and the core units of the first chip are located on the periphery of the first chip, and the core units of the second chip are located on the periphery of the second chip;

a power via connecting one of the power elements of the first chip to one of the core units of the second chip; and

a thermal via connecting one of the core units of the first chip to one of the heat spreader elements of the second chip.

18. The 3D chip stack structure of claim **17**, wherein the first chip is located beneath the second chip.

19. The 3D chip stack structure of claim **17**, wherein the first chip is located above the second chip.

20. The 3D chip stack structure of claim **1**, further comprising a dielectric material structure encasing the first chip, the second chip, the power via and the thermal via.

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